

Introduction to Data Pre-processing

is the process of cleaning, transforming, reducing, and organizing raw data into a suitable format for analysis, machine learning, and Business Intelligence applications.

Raw data collected from different sources is often: Incomplete, Noisy, Inconsistent, Redundant

Poor quality data reduces model accuracy and affects business decisions.

Data preprocessing improves: Data quality, Accuracy, Reliability, Processing efficiency

Objectives of Data Pre-processing

1. Improve data quality
2. Remove errors and inconsistencies
3. Reduce data complexity
4. Prepare data for analytics and ML
5. Improve model performance

Data Validation

Data Validation is the process of checking whether data is correct, complete, and meaningful before processing. Validation ensures that invalid or incorrect data does not enter the system.

Types of Validation

A. Range Validation - Checks whether values lie within a valid range.

Example - Age should be between 1 and 120.

B. Format Validation - Checks proper data format. **Example** - Email format validation.

C. Type Validation - Checks correct datatype.

Example - Phone number should contain numeric values.

Incomplete Data

refers to datasets containing missing or unavailable values.

Causes of Incomplete Data - Data entry errors, Sensor failures, Missing survey responses, System issues

Problems Caused - Reduced model accuracy, Incorrect analysis, Biased predictions

Handling Techniques

A. Remove Missing Records - Delete incomplete rows.

B. Fill Missing Values - Replace missing values using: Mean, Median, Mode

C. Predict Missing Values - Use ML techniques.

Advantages of Handling Incomplete Data

1. Better data quality.
2. Improved analytics accuracy.

Data Affected by Noise

Noise refers to random errors, incorrect values, irrelevant information, or inconsistencies present in data that reduce its accuracy and reliability.

Types of Data Affected by Noise

1. Numerical Data

- Contains incorrect or extreme values (outliers).
- Example: Age recorded as 250 years.

2. Categorical Data

- Contains incorrect category labels or spelling errors.
- Example: "Male", "Mlae", "male".

3. Text Data

- Contains typing mistakes, special characters, and irrelevant words.
- Example: "Machne Learning" instead of "Machine Learning".

4. Sensor Data

- Noise occurs due to measurement errors or environmental factors.
- Example: Incorrect temperature readings from sensors.

5. Time-Series Data

- Contains sudden fluctuations or abnormal spikes.
- Example: Unexpected stock price spikes due to recording errors.

Impact of Noise on Data Quality

1. Reduces Accuracy

Incorrect values lead to inaccurate analysis and predictions.

2. Affects Decision Making

Poor-quality data can result in wrong business decisions.

3. Decreases Model Performance

Machine learning models trained on noisy data may produce poor results.

4. Increases Processing Time

Additional preprocessing is required to clean noisy data.

5. Produces Misleading Insights

Noise can hide actual patterns and trends in the data.

Data Transformation

Data Transformation converts data into a suitable format for analysis and machine learning.

Objectives - Improve consistency, Normalize data, Prepare features for models

Types of Data Transformation - Standardization, Normalization, Aggregation, Encoding

Advantages

1. Better model performance.
2. Improved data compatibility.

Standardization

Standardization transforms data so that features have similar scale with mean 0 and standard deviation 1.

Formula: $Z = (X - \text{Mean}) / \text{Standard Deviation}$

Where:

- **Z** = Standardized value
- **X** = Original value
- **Mean** = Average of all values
- **Standard Deviation** = Measure of data spread

Importance - Different scales affect machine learning algorithms.

Example

Age values: 18, 25, 40

Income values: 10000, 50000, 90000

Standardization converts them into comparable scales.

Advantages

1. Improves ML performance.
2. Faster convergence.

Disadvantages

1. Sensitive to outliers.

Feature Extraction - is the process of selecting or creating important attributes from raw data. Large datasets may contain unnecessary features. Feature extraction reduces complexity while preserving useful information.

Advantages

1. Reduces dimensionality.
2. Improves model accuracy.
3. Faster computation.

Disadvantages - Possible information loss.

Data Reduction

Data Reduction is the process of reducing the volume of data while preserving important information. It helps in efficient storage, processing, and analysis of large datasets.

Significance of Data Reduction

- Reduces storage requirements.
- Decreases computational cost.
- Improves processing speed.
- Simplifies data analysis.
- Removes redundant and unnecessary data.

Types of Data Reduction - Sampling, Feature selection, PCA

Advantages

1. Faster processing.
2. Lower storage cost.

Disadvantages

1. Information loss possible.

Sampling

Sampling is a data reduction technique in which a representative subset of data is selected from a large dataset for analysis.

Types of Sampling

1. Simple Random Sampling

- Every data item has an equal chance of being selected.
- Example: Randomly selecting 100 customers from a database.

2. Systematic Sampling

- Selects every kth record from the dataset.
- Example: Selecting every 10th customer.

3. Stratified Sampling

- Data is divided into groups (strata), and samples are selected from each group.
- Example: Sampling students from different departments.

Feature Selection

Feature Selection identifies the most relevant features for model training. Irrelevant features reduce performance and increase complexity.

Methods of Feature Selection

A. Filter Method - Uses statistical techniques.

B. Wrapper Method - Tests different feature subsets.

C. Embedded Method - Feature selection occurs during training.

Principal Component Analysis (PCA)

Principal Component Analysis (PCA) is a dimensionality reduction technique that transforms high-dimensional data into fewer principal components. PCA reduces complexity while preserving maximum variance.

Working of PCA

Step 1: Standardize data.

Step 2: Compute covariance matrix.

Step 3: Calculate eigenvectors and eigenvalues.

Step 4: Select principal components.

Advantages

1. Reduces dimensionality.
2. Faster processing.
3. Removes redundancy.

Disadvantages

1. Reduced interpretability.
2. Information loss possible.

Feature Selection VS Principal Component Analysis (PCA)

Parameter	Feature Selection	Principal Component Analysis (PCA)
Definition	Selects the most relevant features from the original dataset.	Creates new features (principal components) by combining original features.
Data Transformation	No transformation; original features are retained.	Transforms data into a new feature space.
Interpretability	High, since original features are preserved.	Low, as principal components are combinations of features.
Dimensionality Reduction	Reduces dimensions by removing irrelevant features.	Reduces dimensions by creating fewer principal components.
Information Retention	May lose some useful information if important features are removed.	Retains most of the data variance.
Computational Cost	Generally lower.	Higher due to matrix calculations and eigenvalue computation.
Applicability	Useful when feature importance is required.	Useful when dealing with highly correlated features and large datasets.
Output	Subset of original features.	New transformed components.

Data Discretization

is the process of converting continuous numerical data into a finite number of intervals or categories. It simplifies data and makes it easier to understand and analyze.

Methods of Data Discretization

1. Equal Width Binning

- Divides the data range into intervals of equal size.
- Simple and easy to implement.

Example:

Age (0–100) → 0–20, 21–40, 41–60, 61–80, 81–100.

Effectiveness:

- Easy to understand.
- May produce uneven distribution of data.

2. Equal Frequency Binning

- Divides data so that each bin contains approximately the same number of records.

Example:

100 records divided into 5 bins → 20 records per bin.

Effectiveness:

- Balances data distribution.
- More meaningful than equal-width bins for skewed data.

3. Cluster-Based Discretization

- Uses clustering algorithms such as K-Means to form groups.

Effectiveness:

- Produces natural groupings.
- Captures hidden patterns effectively.
- More computationally expensive.

Advantages of Discretization

- **Simplifies complex numerical data.**
- **Improves visualization and reporting.**
- **Helps identify patterns and trends.**
- **Reduces noise in data.**

Data Exploration

is the process of analyzing datasets to understand their structure, patterns, trends, relationships, and characteristics before applying machine learning or analytics techniques.

Raw data may contain: Missing values, Outliers, Noise, Inconsistencies

Objectives of Data Exploration - Understand dataset distribution & characteristics, Detect outliers and anomalies, Analyze data distribution, Identify patterns and trends, Improve decision-making and model accuracy.

Types of Data Exploration - Univariate Analysis, Bivariate Analysis, Multivariate Analysis

Univariate Analysis

analyzes only one variable or attribute at a time. It helps understand: Distribution, Central tendency Variability, Outliers

Objectives of Univariate Analysis - Summarize data characteristics, Detect unusual values, Understand data distribution, Prepare data for modeling

Univariate Analysis Workflow

Single Variable->Graphical Analysis->Statistical Analysis->Insights and Patterns

1. Graphical Analysis of Categorical Attributes

Graphical analysis of categorical attributes represents categorical data visually to understand frequency and distribution. Categorical attributes contain labels or categories instead of numeric values.

Examples - Gender, Department, Product category

Common Graphs for Categorical Data

A. Bar Chart - represents categorical data using rectangular bars.

Advantages - Easy comparison, Simple visualization.

B. Pie Chart - represents categories as portions of a circle.

Example - Market share distribution.

Advantages

1. Shows percentage contribution.
2. Easy visual understanding.

Disadvantages

1. Difficult with many categories.

C. Count Plot - Displays frequency count of each category.

Applications: Frequency analysis, Classification datasets

2. Graphical Analysis of Numerical Attributes

Graphical analysis of numerical attributes visually represents numerical data distributions and patterns. Numerical attributes contain measurable values.

Examples: Salary, Age, Marks, Temperature

Common Graphs for Numerical Data

A. Histogram

Histogram displays frequency distribution of numerical data using continuous bars.

Example - Marks distribution among students.

Advantages

1. Shows distribution clearly.
2. Detects skewness and spread.

B. Box Plot - represents distribution using quartiles and highlights outliers.

Components - Minimum value, First quartile (Q1), Median, Third quartile (Q3), Maximum value

Advantages

1. Detects outliers easily.
2. Shows spread and skewness.

C. Density Plot - shows smooth probability distribution of data.

Applications - Statistical analysis, ML preprocessing

Measures of Central Tendency for Numerical Attributes

Measures of Central Tendency describe the central or typical value of a dataset.

Types of Central Tendency - Mean, Median, Mode

A. Mean - Mean is the average of all values.

Formula: $\bar{x} = \frac{\sum x}{n}$

- $\sum x$ = Sum of values
- n = Number of observations

Example Values: 10, 20, 30

Mean: $(10 + 20 + 30)/3 = 20$

Advantages

1. Easy to calculate.
2. Uses all observations.

B. Median - Median is the middle value in sorted data.

Example 5, 10, 15, Median = 10

Advantages - Not affected by outliers.

Disadvantages - Does not use all values.

C. Mode - is the most frequently occurring value.

Example 2, 2, 3, 5, Mode = 2

Advantages - Useful for categorical data.

Disadvantages - May not exist uniquely.

Measures of Dispersion for Numerical Attributes

Measures of Dispersion describe how spread out data values are from the central value.

Dispersion helps understand: Variability, Consistency, Data spread

Types of Dispersion Measures - Range, Variance, Standard Deviation, Interquartile Range (IQR)

A. Range - Difference between maximum and minimum values.

Formula: $Range = Maximum - Minimum$

Example - 10, 20, 50, Range = 50 - 10 = 40

B. Variance - measures average squared deviation from the mean.

Formula: $\sigma^2 = \frac{\sum (x - \mu)^2}{N}$

Advantages - Measures variability accurately.

Disadvantages - Difficult interpretation due to squared units.

C. Standard Deviation - measures average spread of data around the mean.

Formula: $\sigma = \sqrt{\frac{\sum (x - \mu)^2}{N}}$

D. Interquartile Range (IQR)

IQR measures spread of middle 50% of data.

Formula : $IQR = Q3 - Q1$

5. Identification of Outliers for Numerical Attributes

Outliers are data values significantly different from other observations.

Causes of Outliers - Data entry errors, Measurement errors, Rare events, Fraudulent activities

Problems Caused by Outliers - Reduced model accuracy, Incorrect analysis, Distorted statistical measures

Methods for Outlier Detection

A. Box Plot Method - Uses quartiles and IQR.

B. Z-Score Method - Detects values far from mean.

Z-Score = $(Value - Mean) / Standard Deviation$

C. Scatter Plot - Visual identification of abnormal values.

Introduction to Bivariate Analysis

is the statistical analysis of two variables to study their relationship, association, dependency, or correlation. Unlike univariate analysis, which studies only one variable, bivariate analysis examines how one variable changes with another.

Objectives of Bivariate Analysis

1. Study relationships between variables

2. Detect positive or negative correlations
3. Compare datasets
4. Support predictive analysis
5. Improve decision-making

Types of Variables in Bivariate Analysis

Variable Type	Example
Numerical vs Numerical	Age and Salary
Categorical vs Categorical	Gender and Product Choice
Numerical vs Categorical	Salary by Department

Graphical Analysis in Bivariate Analysis

Graphical analysis visually represents relationships between two variables using graphs and charts.

Common Graphical Methods

A. Scatter Plot - displays relationships between two numerical variables using points on a graph.

Example - Relationship between: Study hours, Exam marks

Interpretation

Positive Correlation - As one variable increases, the other also increases.

Negative Correlation - As one variable increases, the other decreases.

No Correlation - No clear relationship between variables.

Advantages

1. Easy relationship visualization.
2. Detects trends and outliers.

Disadvantages

1. Difficult for very large datasets.

B. Line Graph - shows relationship and trends between variables over time.

C. Grouped Bar Chart - Compares two categories visually using grouped bars.

Measures of Correlation for Numerical Attributes

Correlation measures the strength and direction of relationship between two numerical variables. Correlation analysis determines whether variables move together or independently.

Types of Correlation

A. Positive Correlation - Both variables increase together.

Example - Experience and salary.

B. Negative Correlation

One variable increases while the other decreases.

Example - Price and demand.

C. Zero Correlation - No relationship between variables.

Contingency Tables for Categorical Attributes

A Contingency Table is a tabular representation used to analyze relationships between two categorical variables. It displays frequency counts of combinations of categories.

Example

	Gender	Likes Product	Dislikes Product
Male	40		10
Female	35		15

Components of Contingency Table

1. Rows represent one categorical variable
2. Columns represent another variable
3. Cells contain frequency counts

Purpose of Contingency Tables

1. Analyze categorical relationships
2. Detect associations
3. Compare category frequencies

Advantages of Bivariate Analysis

1. Identifies variable relationships
2. Supports predictive analysis
3. Improves decision-making
4. Detects dependencies and trends

Disadvantages of Bivariate Analysis

1. Complex interpretation sometimes
2. Correlation does not imply causation

Introduction to Multivariate Analysis

Multivariate Analysis is the statistical analysis of more than two variables simultaneously to understand relationships, patterns, dependencies, and interactions among variables.

Real-world datasets often contain multiple variables that influence each other.

Example: Sales may depend on: Advertising, Product price, Customer income, Season

Multivariate analysis helps study all these variables together.

Objectives of Multivariate Analysis

1. Analyze relationships among multiple variables
2. Identify patterns and trends
3. Reduce data complexity
4. Improve prediction accuracy
5. Support business decision-making

Types of Variables in Multivariate Analysis

Type	Example
Multiple Numerical Variables	Age, Salary, Experience

Type	Example
Numerical + Categorical	Salary, Department
Multiple Categorical Variables	Gender, Product Type, Region

1. Graphical Analysis in Multivariate Analysis

Graphical Analysis in Multivariate Analysis visually represents relationships among multiple variables using advanced visualization techniques.

Common Graphical Techniques

Scatter Plot Matrix

A Scatter Plot Matrix displays pairwise relationships between multiple numerical variables.

Example

Relationship among: Age, Salary, Experience, Education

Advantages

- Shows multiple relationships together.
- Detects correlations and trends.

Disadvantages

- Complex for very large datasets.

Heatmap

A Heatmap uses color intensity to represent correlation or relationship strength among variables.

Advantages

- Easy interpretation.
- Detects strong and weak correlations quickly.

Disadvantages

- Difficult for extremely large datasets.

3D Scatter Plot

Represents relationships among three numerical variables in three-dimensional space.

Advantages

- Better multidimensional visualization.

Disadvantages

- Difficult interpretation sometimes.

Measures of Correlation for Numerical Attributes

Correlation measures the strength and direction of relationships among numerical variables. In multivariate analysis, correlation identifies how variables influence one another.

Types of Correlation

A. Positive Correlation - Variables increase together

B. Negative Correlation - One variable increases while another decreases.

C. Zero Correlatio - No relationship exists between variables.

Correlation Matrix

A Correlation Matrix is a table showing correlation coefficients among multiple numerical variables.

Difference Between Univariate, Bivariate, and Multivariate Analysis

Parameter	Univariate Analysis	Bivariate Analysis	Multivariate Analysis
Meaning	Analysis of one variable	Analysis of two variables	Analysis of three or more variables
Purpose	Understand characteristics of a single variable	Study relationship between two variables	Study relationships among multiple variables
Number of Variables	1	2	More than 2
Focus	Distribution and summary statistics	Correlation or association	Complex interactions and dependencies
Techniques	Mean, Median, Mode, Histogram	Scatter Plot, Correlation, Contingency Table	PCA, Multiple Regression, Cluster Analysis
Graphical Methods	Bar Chart, Pie Chart, Histogram	Scatter Plot, Line Graph	3D Plot, Heat Map, Pair Plot
Complexity	Low	Moderate	High
Example	Analyzing students' marks	Relationship between study hours and marks	Relationship among study hours, attendance, and marks